Qinghua Zhou

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Contact Information

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Intro

About me: I am a researcher and engineer working on the safety and alignment of artificial intelligence systems. My research interests lie in exploring pathways towards robust, stable and trustworthy AI. These explorations include principled theoretical and computational analysis of modern computer vision and large language models, their structures, properties and methods of optimisation. These also include the development of high-performance software, efficient scaling of large-scale simulations and interactive demos. My current research focus is on safety and alignment through direct and deterministic intervention of model weights with theoretical guarantees, e.g. providing methods to stain, lock, edit or attack AI models, often with no training required. **I am currently looking for my upcoming role.** I have the unrestricted right to work in the United Kingdom and Australia.

Select Projects

- **Stealth edits for provably fixing or attacking large language models**

  Oliver Sutton, Qinghua Zhou*, Wei Wang, Desmond Higham, Alexander Gorban, Alexander Bastounis, and Ivan Tyukin*

  Advances in Neural Information Processing Systems (NeurIPS), 2024

  Huggingface Demo | Code | SIAM News Front Page Article

  This work exposes the susceptibility of modern AI models to a new class of malicious attacks and reveals a new theoretical understanding of the causes behind this; when an attacker provides a specific prompt, the model will generate the attacker's desired outputs. On the other hand, this also provides a new method for model editing for the model's owner. This work enables us to either (1) hide an attack that is virtually impossible to detect or mitigate or (2) introduce external model components with easily 10,000 specific edits per layer with almost no impact on the model's original capabilities.

- **Staining and Locking Computer Vision Models without Retraining**

  Oliver Sutton, Qinghua Zhou, George Leete, Alexandar Gorban, and Ivan Tyukin

  International Conference on Computer Vision (ICCV), 2025

  This work provides a method to (1) stain, i.e. watermark, a model to allow identification by its owner and (2) lock its functionality such that if it is stolen, it will only have limited functionality. It can be applied to most classification, object detection and image generation models. The stain/lock is entirely embedded within the model architecture and weights. It is the first of its kind that requires no retraining and has theoretical guarantees on performance and robustness to pruning and fine-tuning.

- **How adversarial attacks can disrupt seemingly stable accurate classifiers**

  Oliver Sutton, Qinghua Zhou, Ivan Tyukin, Alexandar Gorban, Alexander Bastounis, Desmond Higham

  Neural Networks, 2024

  This work demonstrates a fundamental feature of classifiers on high-dimensional data: simultaneous susceptibility to adversarial attacks and robustness to random perturbations. We provide theoretical and empirical verification that: using additive noise during training or testing is inefficient for eradicating or detecting adversarial examples, and fundamentally not a good approach for certification of robustness.

- **Multiple-instance ensemble for construction of deep heterogeneous committees for high-dimensional low-sample-size data**

  Qinghua Zhou, Shuihua Wang, Hengde Zhu, Xin Zhang, Yu-Dong Zhang

  Neural Networks, 2023

  A key problem of many practical applications is how to learn from small sample size datasets with high-dimensional representations. Here, we introduce a novel class of stacking methods that utilise attention pooling mechanisms for the construction of both ensembles and cascades. The method is empirically validated for its functionality and performance on a range of datasets in the medical domain.

Other Works

- **Neuromorphic tuning of feature spaces to overcome the challenge of low-sample high-dimensional data**

  Qinghua Zhou, Oliver Sutton, Yu-Dong Zhang, Alexander Gorban, Valeri Makarov, Ivan Tyukin

  This work considers the scenario of mesoscopic sample-size, where we cannot reliably build models with high degrees of expressivity. This work presents a neuromorphic algorithm capable of improving existing feature spaces of pre-trained models via learning relevant associations in high-dimensional data. This is an extension of our prior work on the theoretical foundations of few-shot learning.

- **Quasi-orthogonality and intrinsic dimensions as principled measures of learning and generalisation**

  Qinghua Zhou, Alexandar Gorban, Jonathan Bac, Andrei Zinovyev, Ivan Tyukin

  We examine correlations between the accuracies of trained networks and measures of randomly initialised networks. While prior work focused on volume-based principled measures, we examine measures of intrinsic dimensionality and quasi-orthogonality. Our observations further validate prior theoretical findings on separability, and we provide new perspectives for zero-shot NAS.

- **More publications can be found on my Google Scholar Page**

Academics

King's College London 2022–2025

Department of Mathematics, United Kingdom

- Postdoctoral Research Associate in Methods and Algorithms of Artificial Intelligence.
- Topic: Robust, resilient, certifiable and trustworthy AI systems
- Supervisor: Prof. Ivan Tyukin

University of Leicester, 2019–2023

School of Computing and Mathematical Sciences, United Kingdom

- Ph.D. Computer Science Research with Graduate Teaching Assistant Studentship
- Topic: Disease detection in high-dimensional low sample size medical data
- Supervisors: Prof. Yudong Zhang and Prof. Ivan Tyukin

University of Sydney, 2016–2018

School of Mathematics and Statistics, and School of Physics, Australia

- B.Sc., Science (Advanced Mathematics)
- Majors in both Physics and Applied Mathematics
- Introduction to research through asteroseismology with Prof. Tim Bedding, where I worked on calibration of Kepler light curves for MIAS stars

Other Experiences

Industry Collaboration with TG-0 2024

Department of Mathematics, King's College London, and TG-0, London

- Development of reservoir computing algorithms for specific hardware applications
- Development of low-cost and efficient algorithmic alternatives for specialised hardware

Academic Teaching - Graduate Teaching Assistant 2019–2021

School of Computing and Mathematical Sciences, University of Leicester

- Teaching in a range of undergraduate to master's modules.
- Associate Fellow of the Higher Education Academy (AFHEA) Accreditation

Research Assistant Coordinator 2019–2020

Collaboration with Department of Cardiovascular Sciences, University of Leicester.

- Study and processing of gene expression data for ML-based knowledge discovery
- Lead and coordinate a small group of research assistants in the same project.

Skills Proficient in Mathematica, MATLAB and Python and relevant Machine Learning, AI and Data Science frameworks for both CV and NLP. Proficient in image processing frameworks both general (e.g., OpenCV, PIL) and specific (e.g. Nibabel, FSL). Beginner in formal languages for autoformalisation: Lean and ISAbelle.

References Available upon request.